

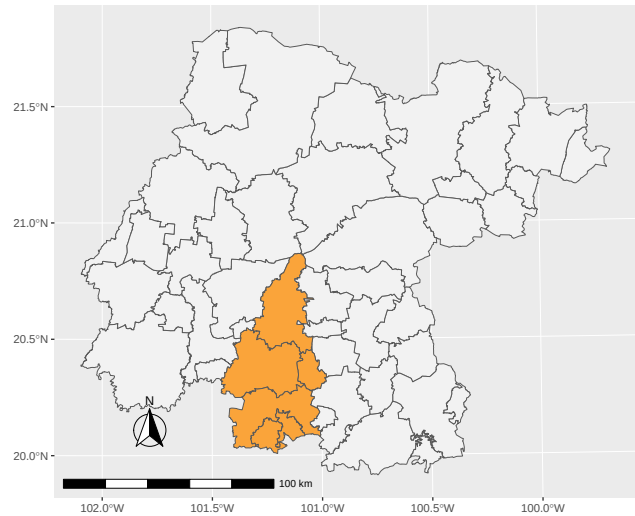
A Additional Figures and Tables

Figure A.1: Maps

(a) Guanajuato



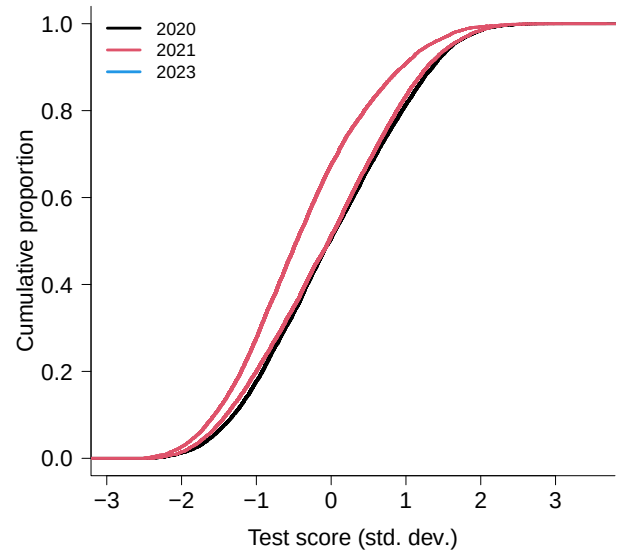
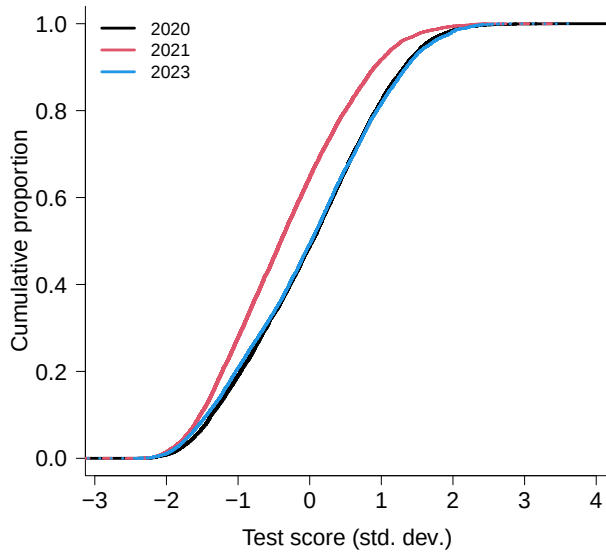
(b) Municipalities where the RIMA assessment took place in 2020



Note: Figure A.1a shows where Guanajuato is located in Mexico (colored). Figure A.1b depicts the administrative division of municipalities in Guanajuato. The six municipalities where the RIMA assessment took place in 2020 (Jaral del Progreso, Moroleón, Salamanca, Uriangato, Valle de Santiago, y Yuriria) are colored.

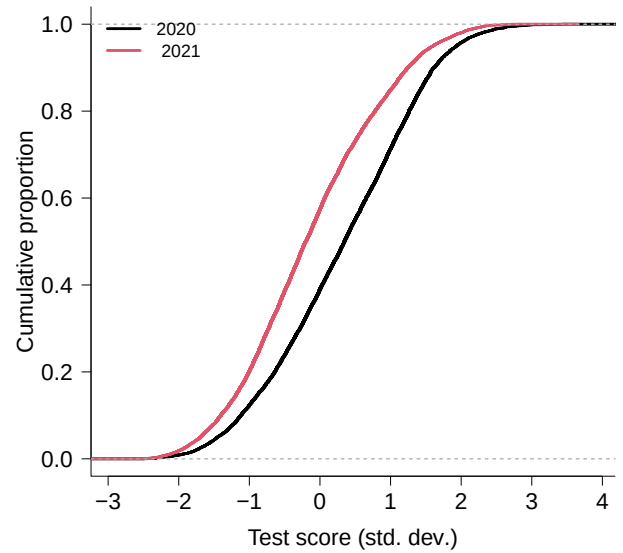
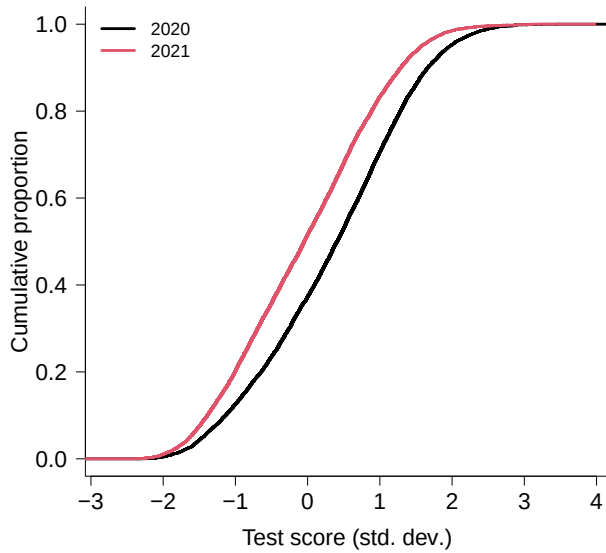
Figure A.2: Cumulative distribution across years
(a) Grade 5 - Spanish

(b) Grade 5 - Math



(c) Grade 6 - Spanish

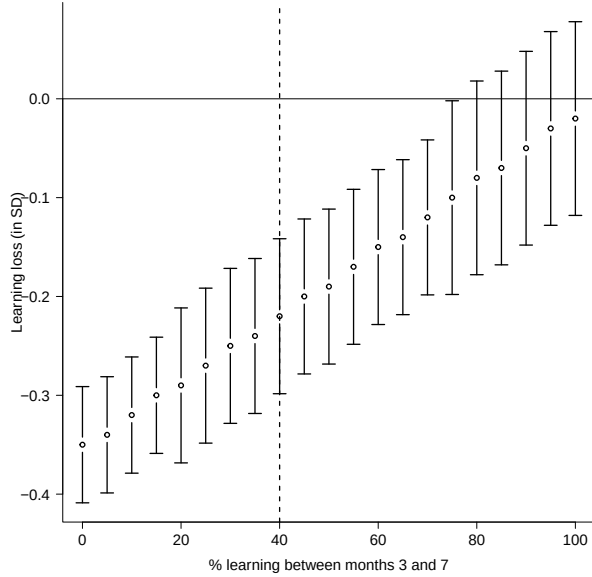
(d) Grade 6 - Math



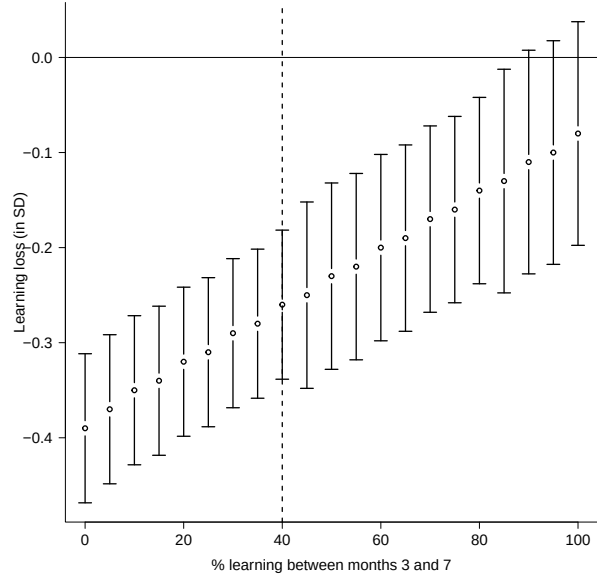
Note: This figure presents the cumulative distribution of test scores across time for students in Grade 5 and 6, in both the mathematics and the Spanish test.

Figure A.3: Learning loss by November 2021 under different assumptions of learning throughout the school year

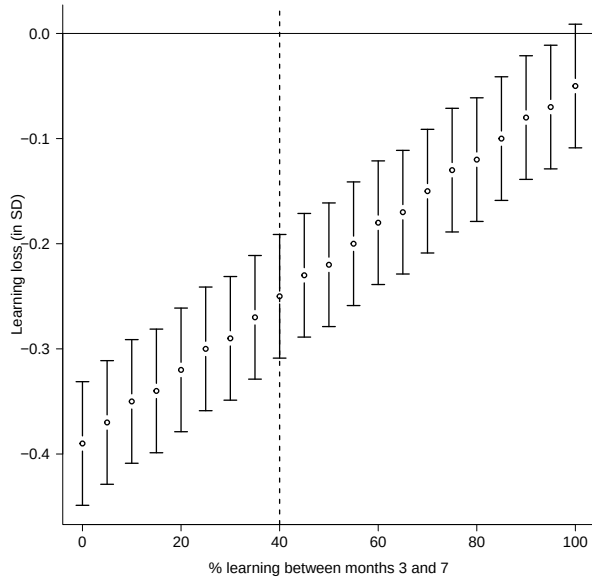
(a) Grade 5 - Spanish



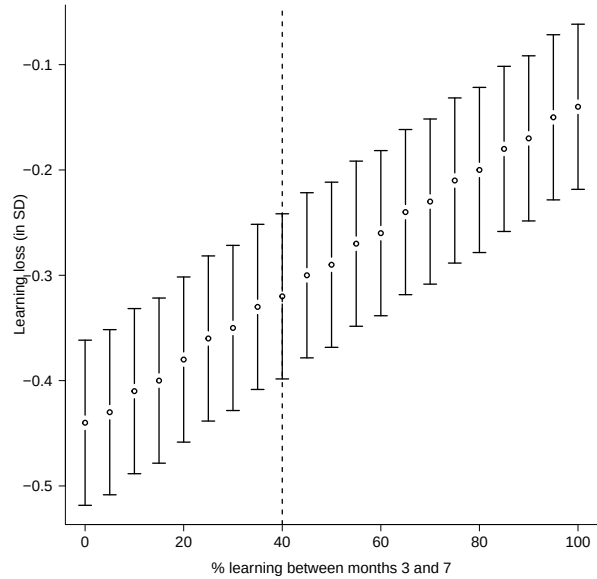
(b) Grade 5 - Math



(c) Grade 6 - Spanish



(d) Grade 6 - Math

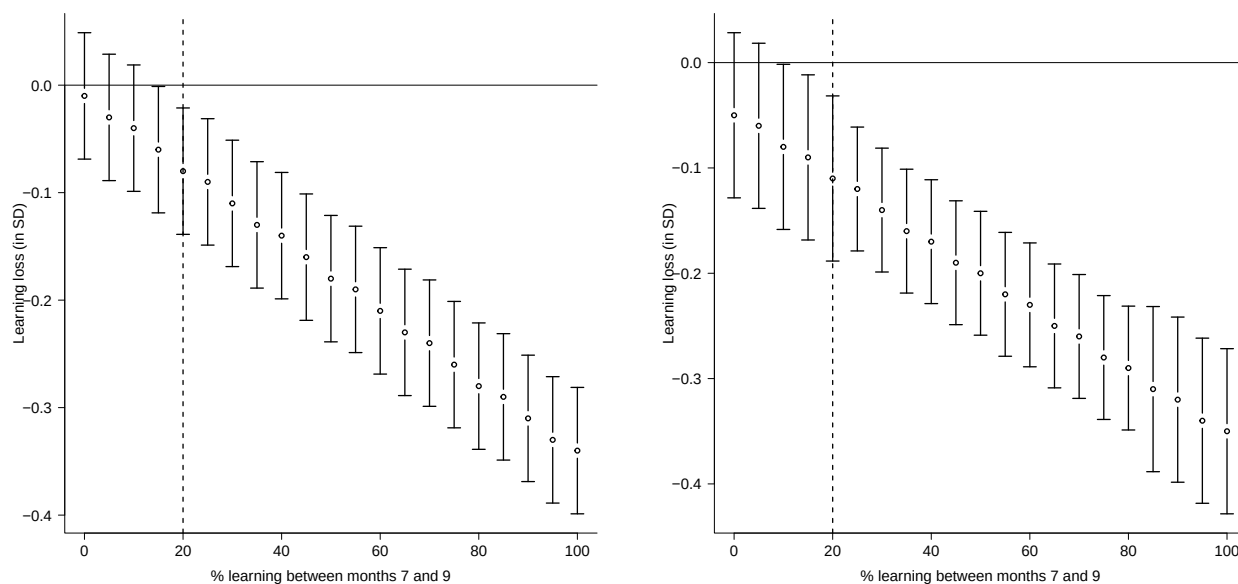


Note: This figure presents the learning loss estimates (y-axis) under different assumptions of how much students learn between the third and the seventh month of the school year (when the first and the second assessment took place) as a proportion of the learning that takes place throughout the academic year (x-axis). The vertical dotted line represents the baseline assumption of linear learning throughout the school year (i.e., that they learn 40% between the third and seventh month of the school, which is equivalent to 40% of the academic year duration, which lasts ten months).

Figure A.4: Learning loss by June 2023 under different assumptions of learning throughout the school year

(a) Grade 5 - Spanish

(b) Grade 5 - Math



Note: This figure presents the learning loss estimates (y-axis) under different assumptions of how much students learn between the seventh and the ninth month of the school year (when the first and the third assessment took place) as a proportion of the learning that takes place throughout the academic year (x-axis). The vertical dotted line represents the baseline assumption of linear learning throughout the school year (i.e., that they learn 20% between the seventh and the ninth month of the school, which is equivalent to 20% of the academic year duration, which lasts ten months).

Table A.1: Municipalities tested in 2020 compared to the rest of the state (using 2021 data)

	(1) Not tested in 2020	(2) Tested in 2020	(3) Difference
Panel A: Grade 5			
Spanish: % correct	39.43 (15.90)	40.66 (15.65)	1.23** (0.49)
Math: % correct	34.42 (14.71)	35.72 (14.83)	1.30** (0.51)
Spanish: IRT	-0.42 (0.94)	-0.35 (0.92)	0.07** (0.03)
Math: IRT	-0.46 (0.95)	-0.38 (0.95)	0.07** (0.03)
Female (%)	49.09 (49.99)	49.55 (50.00)	0.46 (0.49)
Public (%)	90.91 (28.75)	94.39 (23.01)	3.48*** (1.26)
Rural (%)	31.11 (46.29)	31.80 (46.57)	0.69 (2.76)
Marginalized (%)	1.48 (12.08)	0.00 (0.00)	-1.48*** (0.18)
Panel B: Grade 6			
Spanish: % correct	44.00 (17.08)	45.93 (17.48)	1.93*** (0.58)
Math: % correct	37.08 (15.41)	39.24 (16.49)	2.16*** (0.64)
Spanish: IRT	-0.15 (0.99)	-0.05 (1.01)	0.10*** (0.03)
Math: IRT	-0.25 (0.95)	-0.13 (1.00)	0.12*** (0.04)
Female (%)	49.41 (50.00)	49.52 (50.00)	0.11 (0.50)
Public (%)	90.90 (28.77)	94.27 (23.23)	3.38*** (1.28)
Rural (%)	31.17 (46.32)	32.30 (46.76)	1.13 (2.77)
Marginalized (%)	1.57 (12.45)	0.00 (0.00)	-1.57*** (0.19)

Notes: This table presents data for 2021 comparing municipalities that were not tested in 2020 (Column 1) to those tested in 2020 (Column 2). The difference (and its standard error in parenthesis) is presented in Column 3. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.2: Municipalities tested in 2020 compared to the rest of the state (using 2023 data)

	(1) Not tested in 2020	(2) Tested in 2023	(3) Difference
Panel A: Grade 5			
Spanish: % correct	51.09 (22.59)	53.33 (22.23)	2.23*** (0.66)
Math: % correct	44.91 (20.27)	47.34 (20.25)	2.42*** (0.64)
Spanish: IRT	-0.10 (1.04)	-0.01 (1.03)	0.09*** (0.03)
Math: IRT	-0.16 (1.02)	-0.05 (1.01)	0.11*** (0.03)
Female (%)	49.12 (49.99)	49.14 (49.99)	0.02 (0.49)
Public (%)	90.48 (29.35)	93.68 (24.33)	3.20** (1.37)
Rural (%)	31.32 (46.38)	31.85 (46.59)	0.53 (2.80)
Marginalized (%)	1.58 (12.47)	0.00 (0.00)	-1.58*** (0.20)

Notes: This table presents data for 2023 comparing municipalities that were not tested in 2020 (Column 1) to those tested in 2020 (Column 2). The difference (and its standard error in parenthesis) is presented in Column 3. We only present data for Grade 5 students since Grade 6 students are not part of our main estimating sample. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.3: Learning loss in Spanish and mathematics in six municipalities (all students) using the % of correct answers as the outcome

	Spanish				Math			
	All		Common		All		Common	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Grade 6 (α_1)	5.9***	5.8***	8***	7.9***	5.3***	5.3***	7.2***	7.1***
	(.56)	(.55)	(.69)	(.69)	(.66)	(.66)	(.81)	(.8)
2021 (α_2)	-5.7***	-5.8***	-6.7***	-6.9***	-5.4***	-5.6***	-6.7***	-6.9***
	(.57)	(.56)	(.78)	(.77)	(.62)	(.63)	(.79)	(.8)
Grade 6 $\times$ 2021 (α_3)	-.61	-.61	-.79	-.75	-1.8**	-1.8**	-2.7***	-2.7***
	(.67)	(.66)	(.87)	(.86)	(.84)	(.84)	(1)	(1)
2023	7.9***	7.7***	1.8**	1.6**	6.9***	6.7***	.078	-.063
	(.63)	(.62)	(.77)	(.75)	(.71)	(.7)	(.85)	(.85)
Grade 5: Average in 2020 (α_0)	50.33	50.33	56.05	56.05	44.60	44.60	44.62	44.62
Grade 5: Learning loss 2021 ($\alpha_2 + 0.4\alpha_1$)	-3.33	-3.51	-3.49	-3.70	-3.28	-3.45	-3.83	-4.03
p-value(H_0 :Learning loss grade 5 (2021)=0)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Grade 5: Learning loss 2023 ($\alpha_4 - 0.2\alpha_1$)	6.70	6.56	0.15	-0.03	5.80	5.68	-1.36	-1.49
p-value(H_0 :Learning loss grade 5 (2023)=0)	0.00	0.00	0.84	0.96	0.00	0.00	0.08	0.06
Grade 6: Learning loss 2021 ($\alpha_2 + \alpha_3 + 0.4\alpha_1$)	-3.94	-4.12	-4.28	-4.45	-5.07	-5.26	-6.57	-6.75
p-value(H_0 :Learning loss grade 6 (2021)=0)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EYOS Grade 5 in 2021 ($\frac{\alpha_2 + 0.4\alpha_1}{\alpha_1}$)	-0.57	-0.61	-0.43	-0.47	-0.62	-0.66	-0.53	-0.57
p-value(H_0 :EYOS Grade 5 in 2021=0)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EYOS Grade 5 in 2023 ($\frac{\alpha_4 - 0.2\alpha_1}{\alpha_1}$)	1.14	1.13	0.02	-0.00	1.10	1.08	-0.19	-0.21
p-value(H_0 :EYOS Grade 5 in 2023=0)	0.00	0.00	0.84	0.96	0.00	0.00	0.11	0.08
EYOS Grade 6 in 2021 ($\frac{\alpha_2 + \alpha_3 + 0.4\alpha_1}{\alpha_1}$)	-0.67	-0.71	-0.53	-0.56	-0.96	-1.00	-0.91	-0.95
p-value(H_0 :EYOS Grade 6 in 2021=0)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N. of obs.	52,813	52,811	52,793	52,791	52,813	52,811	52,333	52,331
School fixed effects	No	Yes	No	Yes	No	Yes	No	Yes

Notes: This table presents the estimates from Equation 1 using data from the six municipalities tested in 2020, 2021, and 2023. Learning levels are estimated using the percentage of correct answers. Columns 1, 2, 5, and 6 include all questions, while Columns 3, 4, 7, and 8 only include the common items across all grades and years. The learning loss measures are computed as the difference between the actual learning level and the expected level, taking into account that the exam was administered at different points in the school year. The “equivalent years of schooling” (EYOS) indicates the ratio of the learning loss to the normal progression in a typical school year. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.4: Learning loss in Spanish and mathematics using only common items across rounds and grades to estimate the IRT model

	Spanish		Math	
	(1)	(2)	(3)	(4)
Grade 6 (α_1)	.31*** (.035)	.3*** (.035)	.33*** (.028)	.33*** (.027)
2021 (α_2)	-.3*** (.034)	-.3*** (.034)	-.29*** (.031)	-.29*** (.031)
Grade 6 $\times$ 2021 (α_3)	-.1** (.045)	-.1** (.045)	-.044 (.035)	-.043 (.034)
2023	-.0027 (.037)	-.0091 (.037)	.027 (.031)	.019 (.03)
Grade 5: Average in 2020 (α_0)	-0.00	-0.00	0.01	0.01
Grade 5: Learning loss 2021 ($\alpha_2 + 0.4\alpha_1$)	-0.17	-0.18	-0.16	-0.16
p-value(H_0 :Learning loss grade 5 (2021)=0)	0.00	0.00	0.00	0.00
Grade 5: Learning loss 2023 ($\alpha_4 - 0.2\alpha_1$)	-0.06	-0.07	-0.04	-0.05
p-value(H_0 :Learning loss grade 5 (2023)=0)	0.06	0.04	0.18	0.10
Grade 6: Learning loss 2021 ($\alpha_2 + \alpha_3 + 0.4\alpha_1$)	-0.28	-0.29	-0.20	-0.21
p-value(H_0 :Learning loss grade 6 (2021)=0)	0.00	0.00	0.00	0.00
EYOS Grade 5 in 2021 ($\frac{\alpha_2+0.4\alpha_1}{\alpha_1}$)	-0.57	-0.60	-0.47	-0.50
p-value(H_0 :EYOS Grade 5 in 2021=0)	0.00	0.00	0.00	0.00
EYOS Grade 5 in 2023 ($\frac{\alpha_4-0.2\alpha_1}{\alpha_1}$)	-0.21	-0.23	-0.12	-0.14
p-value(H_0 :EYOS Grade 5 in 2023=0)	0.08	0.06	0.20	0.12
EYOS Grade 6 in 2021 ($\frac{\alpha_2+\alpha_3+0.4\alpha_1}{\alpha_1}$)	-0.91	-0.94	-0.60	-0.63
p-value(H_0 :EYOS Grade 6 in 2021=0)	0.00	0.00	0.00	0.00
N. of obs.	52,333	52,331	52,793	52,791
School fixed effects	No	Yes	No	Yes

Notes: This table presents the estimates from Equation 1 using data from the six municipalities tested in 2020, 2021, and 2023. Learning levels are estimated using a single IRT model across grades and years but restricted to the questions asked across all grades and rounds (15 in total). The learning loss measures are computed as the difference between the actual learning level and the expected level, taking into account that the exam was administered at different points in the school year. The “equivalent years of schooling” (EYOS) indicates the ratio of the learning loss to the normal progression in a typical school year. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.5: Learning loss by gender

	Spanish		Math	
	% Correct	IRT	% Correct	IRT
	(1)	(2)	(3)	(4)
Grade 6	.32*** (.032)	.31*** (.032)	.31*** (.038)	.31*** (.037)
Post	-.29*** (.035)	-.31*** (.034)	-.33*** (.037)	-.35*** (.036)
Grade 6 × Post	-.057 (.039)	-.061 (.037)	-.067 (.047)	-.065 (.046)
Female	.27*** (.021)	.26*** (.019)	.11*** (.02)	.099*** (.019)
Grade 6 × Female	.041 (.029)	.041 (.027)	-.016 (.029)	-.0066 (.027)
Post × Female	-.12*** (.027)	-.11*** (.025)	-.12*** (.028)	-.096*** (.025)
Grade 6 × Post × Female	.047 (.041)	.056 (.038)	.021 (.042)	.016 (.039)
2023	.036 (.035)	.027 (.033)	-.016 (.039)	-.029 (.038)
2023 × Female	-.091*** (.027)	-.09*** (.025)	-.064** (.027)	-.057** (.026)
Grade 5: Average in 2020	-0.00	-0.00	-0.02	-0.02
Grade 5: $Loss_1(2021)$ (if Female=1)	-0.27	-0.27	-0.33	-0.32
Grade 5: p-value($H_0 : Loss_1(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 5: $Loss_0(2021)$ (if Female=0)	-0.17	-0.18	-0.20	-0.22
Grade 5: p-value($H_0 : Loss_0(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 5: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.00	0.00	0.00	0.00
Grade 5: $Loss_1(2023)$ (if Female=1)	-0.13	-0.13	-0.14	-0.14
Grade 5: p-value($H_0 : Loss_1(2023) = 0$)	0.00	0.00	0.00	0.00
Grade 5: $Loss_0(2023)$ (if Female=0)	-0.03	-0.04	-0.08	-0.09
Grade 5: p-value($H_0 : Loss_0(2023) = 0$)	0.41	0.25	0.03	0.01
p-value($H_0 : Grade5Loss_1(2023) = Loss_0(2023)$)	0.00	0.00	0.02	0.02
Grade 6: $Loss_1(2021)$ (if Female=1)	-0.28	-0.28	-0.37	-0.37
Grade 6: p-value($H_0 : Loss_1(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: $Loss_0(2021)$ (if Female=0)	-0.22	-0.24	-0.27	-0.29
Grade 6: p-value($H_0 : Loss_0(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.02	0.10	0.00	0.00
N. of obs.	52,813	52,811	52,813	52,811

Notes: This table presents the estimates from an extended version of Equation 1 that includes interactions with a dummy variable for gender using data from the six municipalities tested in 2020 and 2021. The “equivalent years of schooling” (EYOS) shows the ratio of the learning loss to the normal progression in a typical school year for Grade 6 students of both genders. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.6: Learning loss in multigrade vs “monograde” school

	Spanish		Math	
	% Correct	IRT	% Correct	IRT
	(1)	(2)	(3)	(4)
Grade 6	.34*** (.033)	.34*** (.032)	.3*** (.039)	.3*** (.038)
Post	-.35*** (.035)	-.36*** (.034)	-.38*** (.039)	-.39*** (.039)
Grade 6 × Post	-.029 (.04)	-.028 (.04)	-.048 (.05)	-.047 (.05)
Multigrade	-.22*** (.078)		-.12 (.085)	
Grade 6 × Multigrade	-.084 (.068)	-.11* (.061)	-.013 (.07)	-.04 (.063)
Post × Multigrade	-.041 (.086)	-.083 (.084)	-.066 (.1)	-.1 (.1)
Grade 6 × Post × Multigrade	-.0047 (.092)	-.003 (.084)	-.093 (.095)	-.1 (.089)
2023	.009 (.035)	-.00042 (.034)	-.036 (.04)	-.042 (.04)
2023 × Multigrade	-.32*** (.088)	-.34*** (.079)	-.25** (.099)	-.28*** (.093)
Grade 5: Average in 2020	-0.01	-0.01	-0.02	-0.02
Grade 5: $Loss_1(2021)$ (if Multigrade=1)	-0.29	-0.35	-0.33	-0.39
Grade 5: p-value($H_0 : Loss_1(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 5: $Loss_0(2021)$ (if Multigrade=0)	-0.21	-0.22	-0.26	-0.27
Grade 5: p-value($H_0 : Loss_0(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 5: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.45	0.19	0.53	0.29
Grade 5: $Loss_1(2023)$ (if Multigrade=1)	-0.36	-0.38	-0.34	-0.37
Grade 5: p-value($H_0 : Loss_1(2023) = 0$)	0.00	0.00	0.00	0.00
Grade 5: $Loss_0(2023)$ (if Multigrade=0)	-0.06	-0.07	-0.10	-0.10
Grade 5: p-value($H_0 : Loss_0(2023) = 0$)	0.07	0.03	0.01	0.01
p-value($H_0 : Grade5Loss_1(2023) = Loss_0(2023)$)	0.00	0.00	0.01	0.00
Grade 6: $Loss_1(2021)$ (if Multigrade=1)	-0.32	-0.38	-0.47	-0.54
Grade 6: p-value($H_0 : Loss_1(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: $Loss_0(2021)$ (if Multigrade=0)	-0.24	-0.25	-0.31	-0.32
Grade 6: p-value($H_0 : Loss_0(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.36	0.12	0.08	0.01
N. of obs.	52,476	52,474	52,476	52,474

Notes: This table presents the estimates from an extended version of Equation 1 that includes interactions with a dummy variable for whether the school has multigrades (i.e., children from different grades sit in the same classroom and are taught by the same teacher simultaneously) using data from the six municipalities tested in 2020 and 2021. The “equivalent years of schooling” (EYOS) shows the ratio of the learning loss to the normal progression in a typical school year for Grade 6 students in multigrade and monograde schools. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.7: Learning loss in public vs private schools

	Spanish		Math	
	% Correct	IRT	% Correct	IRT
	(1)	(2)	(3)	(4)
Grade 6	.27*** (.089)	.29*** (.089)	.25*** (.09)	.28*** (.091)
Post	-.36*** (.11)	-.36*** (.11)	-.35*** (.097)	-.37*** (.093)
Grade 6 $\times$ Post	.013 (.13)	-.0015 (.12)	-.07 (.12)	-.098 (.11)
Public	-.62*** (.1)		-.37*** (.099)	
Grade 6 $\times$ Public	.068 (.095)	.04 (.094)	.059 (.097)	.022 (.098)
Post $\times$ Public	.011 (.12)	-.00052 (.11)	-.032 (.1)	-.027 (.1)
Grade 6 $\times$ Post $\times$ Public	-.051 (.13)	-.032 (.12)	.013 (.13)	.044 (.12)
2023	.0034 (.11)	.025 (.11)	.027 (.1)	.031 (.1)
2023 $\times$ Public	-.016 (.11)	-.046 (.11)	-.082 (.11)	-.094 (.11)
Grade 5: Average in 2020	-0.00	-0.00	-0.02	-0.02
Grade 5: $Loss_1(2021)$ (if Public=1)	-0.21	-0.23	-0.26	-0.27
Grade 5: p-value($H_0 : Loss_1(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 5: $Loss_0(2021)$ (if Public=0)	-0.25	-0.24	-0.25	-0.25
Grade 5: p-value($H_0 : Loss_0(2021) = 0$)	0.07	0.06	0.04	0.02
Grade 5: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.79	0.91	0.95	0.88
Grade 5: $Loss_1(2023)$ (if Public=1)	-0.08	-0.09	-0.12	-0.12
Grade 5: p-value($H_0 : Loss_1(2023) = 0$)	0.01	0.00	0.00	0.00
Grade 5: $Loss_0(2023)$ (if Public=0)	-0.05	-0.03	-0.02	-0.03
Grade 5: p-value($H_0 : Loss_0(2023) = 0$)	0.60	0.73	0.81	0.80
p-value($H_0 : Grade5Loss_1(2023) = Loss_0(2023)$)	0.77	0.60	0.37	0.35
Grade 6: $Loss_1(2021)$ (if Public=1)	-0.25	-0.26	-0.32	-0.33
Grade 6: p-value($H_0 : Loss_1(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: $Loss_0(2021)$ (if Public=0)	-0.24	-0.24	-0.32	-0.35
Grade 6: p-value($H_0 : Loss_0(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.88	0.85	0.96	0.76
N. of obs.	52,813	52,811	52,813	52,811

Notes: This table presents the estimates from an extended version of Equation 1 that includes interactions with a dummy variable for whether the school is public using data from the six municipalities tested in 2020 and 2021. The “equivalent years of schooling” (EYOS) shows the ratio of the learning loss to the normal progression in a typical school year for Grade 6 students in private and public schools. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.8: Learning loss in urban vs rural schools

	Spanish		Math	
	% Correct	IRT	% Correct	IRT
	(1)	(2)	(3)	(4)
Grade 6	.37*** (.036)	.36*** (.036)	.34*** (.044)	.34*** (.044)
Post	-.39*** (.04)	-.39*** (.039)	-.41*** (.046)	-.41*** (.046)
Grade 6 $\times$ Post	.013 (.047)	.012 (.046)	-.014 (.059)	-.018 (.059)
Rural	-.085 (.064)		.037 (.069)	
Grade 6 $\times$ Rural	-.1 (.067)	-.1 (.066)	-.1 (.078)	-.11 (.077)
Post $\times$ Rural	.1 (.07)	.086 (.07)	.076 (.075)	.055 (.075)
Grade 6 $\times$ Post $\times$ Rural	-.13* (.079)	-.13* (.078)	-.12 (.097)	-.11 (.097)
2023	.027 (.04)	.016 (.039)	-.018 (.049)	-.024 (.048)
2023 $\times$ Rural	-.11 (.07)	-.11* (.068)	-.1 (.076)	-.11 (.075)
Grade 5: Average in 2020	-0.00	-0.00	-0.02	-0.02
Grade 5: $Loss_1(2021)$ (if Rural=1)	-0.18	-0.20	-0.24	-0.27
Grade 5: p-value($H_0 : Loss_1(2021) = 0$)	0.01	0.00	0.00	0.00
Grade 5: $Loss_0(2021)$ (if Rural=0)	-0.24	-0.24	-0.28	-0.28
Grade 5: p-value($H_0 : Loss_0(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 5: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.46	0.58	0.71	0.91
Grade 5: $Loss_1(2023)$ (if Rural=1)	-0.14	-0.15	-0.17	-0.18
Grade 5: p-value($H_0 : Loss_1(2023) = 0$)	0.01	0.00	0.00	0.00
Grade 5: $Loss_0(2023)$ (if Rural=0)	-0.05	-0.06	-0.09	-0.09
Grade 5: p-value($H_0 : Loss_0(2023) = 0$)	0.22	0.13	0.06	0.04
p-value($H_0 : Grade5Loss_1(2023) = Loss_0(2023)$)	0.15	0.15	0.26	0.21
Grade 6: $Loss_1(2021)$ (if Rural=1)	-0.30	-0.32	-0.38	-0.40
Grade 6: p-value($H_0 : Loss_1(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: $Loss_0(2021)$ (if Rural=0)	-0.23	-0.23	-0.29	-0.30
Grade 6: p-value($H_0 : Loss_0(2021) = 0$)	0.00	0.00	0.00	0.00
Grade 6: p-value($H_0 : Loss_1(2021) = Loss_0(2021)$)	0.26	0.17	0.21	0.14
N. of obs.	52,766	52,764	52,766	52,764

Notes: This table presents the estimates from an extended version of Equation 1 that includes interactions with a dummy variable for whether the school is in a rural area using data from the six municipalities tested in 2020 and 2021. The “equivalent years of schooling” (EYOS) shows the ratio of the learning loss to the normal progression in a typical school year for Grade 6 students in rural and urban schools. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.9: Student performance and drop-out

	(1) Not enrolled in 2021	(2) Enrolled in 2021	(3) Difference
Panel A: Grade 5 in 2020			
Spanish: % correct	41.57 (17.91)	46.87 (17.79)	5.30*** (0.97)
Math: % correct	38.08 (16.94)	41.46 (17.05)	3.38*** (0.93)
Spanish: IRT	-0.26 (1.00)	0.03 (1.00)	0.29*** (0.05)
Math: IRT	-0.18 (1.02)	0.02 (1.00)	0.20*** (0.05)
N. of students	996	9,591	
Panel B: Grade 6 in 2020			
Spanish: % correct	44.37 (18.99)	52.84 (18.76)	8.47*** (1.11)
Math: % correct	39.88 (17.74)	46.93 (18.62)	7.05*** (1.09)
Spanish: IRT	-0.12 (1.08)	0.37 (1.04)	0.49*** (0.06)
Math: IRT	-0.10 (1.07)	0.34 (1.03)	0.44*** (0.06)
N. of students	782	10,189	

Notes: This table shows information for students tested in 2020. Column 1 presents the average performance in the standardized test in 2020 of students who are no longer enrolled in a school in Guanajuato in 2021, Column 2 presents the performance of students who are still enrolled in a school in Guanajuato in 2021, and Column 3 presents the difference between the two. Standard deviations (for Columns 1 and 2) and standard errors (for Column 3) are presented in parenthesis. Panel A is for students enrolled in Grade 5 in March 2020, who are expected to be enrolled in Grade 7 in November of 2021. Panel B is for students enrolled in Grade 6 in March of 2020, who are expected to be enrolled in Grade 8 in November of 2021. Standard errors are clustered at the school level (using the 2020 enrollment). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.10: Bounds for learning loss from Lee (2009) style trimming within each school

	Spanish		Math	
	(1)	(2)	(3)	(4)
Panel A: Lower bound (right-tail trimming)				
Grade 6 (α_1)	.34*** (.032)	.33*** (.031)	.31*** (.038)	.3*** (.037)
2021 (α_2)	-.26*** (.033)	-.27*** (.033)	-.3*** (.037)	-.31*** (.037)
Grade 6 $\times$ 2021 (α_3)	-.034 (.039)	-.033 (.038)	-.057 (.048)	-.058 (.048)
2023	-.095*** (.034)	-.1*** (.033)	-.13*** (.039)	-.14*** (.039)
Grade 5: Average in 2020 (α_0)	-0.14	-0.14	-0.15	-0.15
Grade 5: Learning loss 2021 ($\alpha_2 + 0.4\alpha_1$)	-0.13	-0.13	-0.18	-0.18
p-value(H_0 :Learning loss grade 5 (2021)=0)	0.00	0.00	0.00	0.00
Grade 5: Learning loss 2023 ($\alpha_4 - 0.2\alpha_1$)	-0.16	-0.17	-0.20	-0.20
p-value(H_0 :Learning loss grade 5 (2023)=0)	0.00	0.00	0.00	0.00
Grade 6: Learning loss 2021 ($\alpha_2 + \alpha_3 + 0.4\alpha_1$)	-0.16	-0.17	-0.23	-0.24
p-value(H_0 :Learning loss grade 6 (2021)=0)	0.00	0.00	0.00	0.00
EYOS Grade 5 in 2021 ($\frac{\alpha_2 + 0.4\alpha_1}{\alpha_1}$)	-0.38	-0.41	-0.58	-0.61
p-value(H_0 :EYOS Grade 5 in 2021=0)	0.01	0.00	0.00	0.00
EYOS Grade 5 in 2023 ($\frac{\alpha_4 - 0.2\alpha_1}{\alpha_1}$)	-0.48	-0.51	-0.64	-0.67
p-value(H_0 :EYOS Grade 5 in 2023=0)	0.00	0.00	0.00	0.00
EYOS Grade 6 in 2021 ($\frac{\alpha_2 + \alpha_3 + 0.4\alpha_1}{\alpha_1}$)	-0.48	-0.51	-0.76	-0.80
p-value(H_0 :EYOS Grade 6 in 2021=0)	0.00	0.00	0.00	0.00
N. of obs.	50,205	50,204	50,205	50,204
School fixed effects	No	Yes	No	Yes
Panel B: Upper bound (left-tail trimming)				
Grade 6 (α_1)	.34*** (.031)	.34*** (.03)	.31*** (.037)	.31*** (.036)
2021 (α_2)	-.39*** (.033)	-.4*** (.033)	-.43*** (.037)	-.43*** (.037)
Grade 6 $\times$ 2021 (α_3)	-.037 (.038)	-.037 (.038)	-.06 (.047)	-.061 (.047)
2023	.078** (.034)	.07** (.032)	.037 (.039)	.029 (.039)
Grade 5: Average in 2020 (α_0)	0.08	0.08	0.06	0.06
Grade 5: Learning loss 2021 ($\alpha_2 + 0.4\alpha_1$)	-0.26	-0.27	-0.30	-0.31
p-value(H_0 :Learning loss grade 5 (2021)=0)	0.00	0.00	0.00	0.00
Grade 5: Learning loss 2023 ($\alpha_4 - 0.2\alpha_1$)	0.01	0.00	-0.02	-0.03
p-value(H_0 :Learning loss grade 5 (2023)=0)	0.75	0.92	0.49	0.37
Grade 6: Learning loss 2021 ($\alpha_2 + \alpha_3 + 0.4\alpha_1$)	-0.29	-0.30	-0.36	-0.37
p-value(H_0 :Learning loss grade 6 (2021)=0)	0.00	0.00	0.00	0.00
EYOS Grade 5 in 2021 ($\frac{\alpha_2 + 0.4\alpha_1}{\alpha_1}$)	-0.76	-0.79	-0.98	-1.02
p-value(H_0 :EYOS Grade 5 in 2021=0)	0.00	0.00	0.00	0.00
EYOS Grade 5 in 2023 ($\frac{\alpha_4 - 0.2\alpha_1}{\alpha_1}$)	0.03	0.01	-0.08	-0.10
p-value(H_0 :EYOS Grade 5 in 2023=0)	0.75	0.92	0.50	0.39
EYOS Grade 6 in 2021 ($\frac{\alpha_2 + \alpha_3 + 0.4\alpha_1}{\alpha_1}$)	-0.87	-0.90	-1.17	-1.22
p-value(H_0 :EYOS Grade 6 in 2021=0)	0.00	0.00	0.00	0.00
N. of obs.	51,462	51,460	51,462	51,460
School fixed effects	No	Yes	No	Yes

Notes: This table presents the estimates from Equation 1 using data from the six municipalities tested in 2020, 2021, and 2023 using the trimming proposed in Lee (2009). Panel A trims the upper tails of the 2020 and 2023 data — within each school — by the differential testing rate between years (see Table 1). Panel B trims the lower tails of the 2020 and 2023 data — within each school — by the differential testing rates. The learning loss measures are computed as the difference between the actual learning level and the expected level, taking into account that the exam was administered at different points in the school year. The “equivalent years of schooling” (EYOS) indicates the ratio of the learning loss to the normal progression in a typical school year. Standard errors are clustered at the school level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

B Psychometric analysis of the test

A series of psychometric analyses were performed to ensure that the Spanish and mathematics tests used in this study were valid and reliable measures of student ability, that is, that their scores could be interpreted as reflective of student learning, and that differences in scores truly reflect differences in their learning levels.

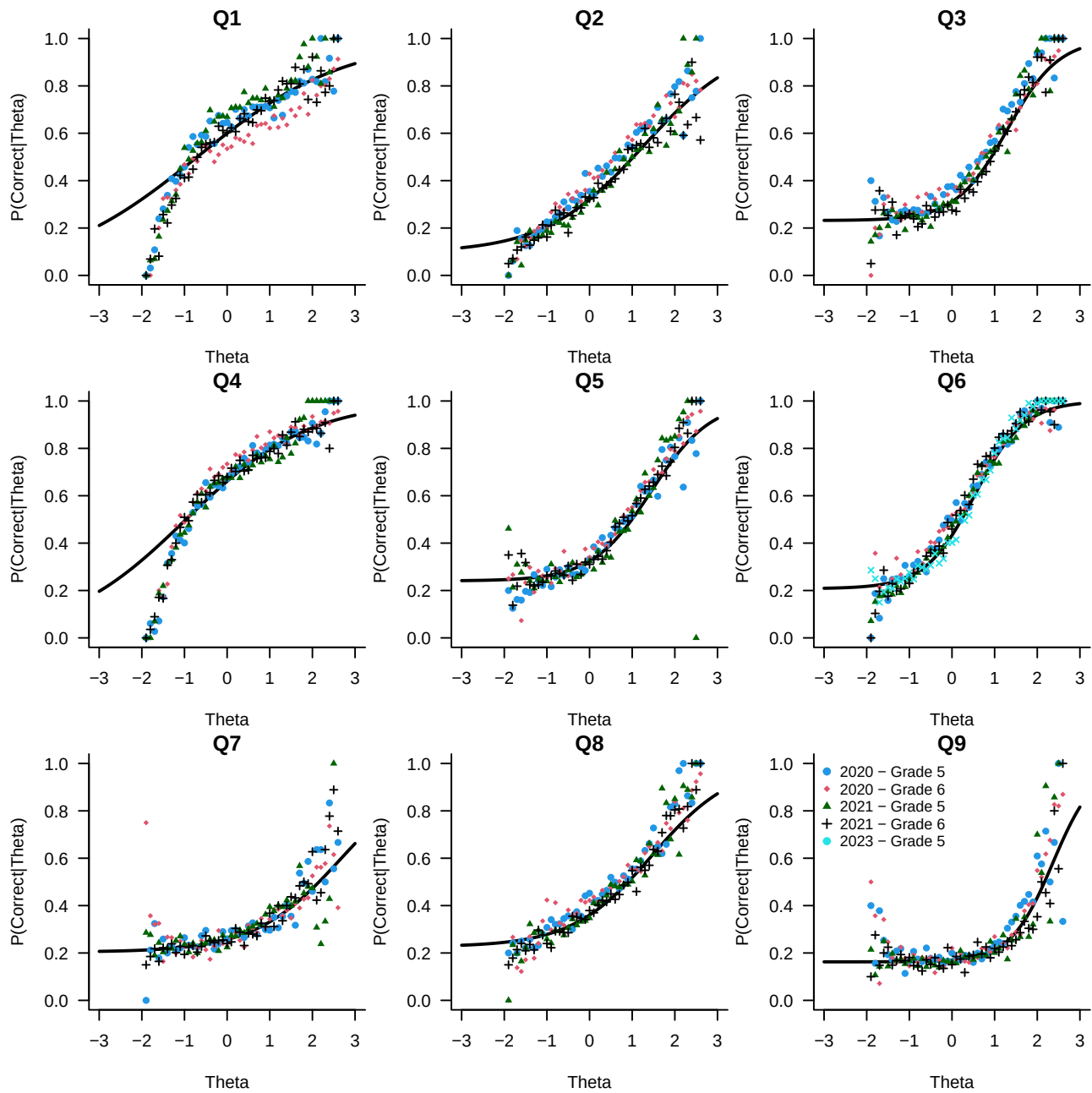
Both item response theory (IRT) and the classical test theory were used to analyze the psychometric properties of the Spanish and mathematics tests. This is the standard procedure in the most advanced learning assessment programs with national coverage (e.g., Chile, Colombia, USA) and international coverage (e.g., PISA, TIMSS, ERCE).

For this study, the Spanish and mathematics test scores were computed using a three-parameter logistic model (3PL). All estimations were made using the multidimensional (MIRT) package in R (Chalmers, 2012). Test scores were computed after removing six questions from the 2019 and 2021 tests with poor psychometric characteristics. For example, three of these questions had average response rates of less than 19 percent, while answering at random should yield a response rate of 25 percent. For the other three, the likelihood of answering the question correctly did not change with the student's ability. Final test scores were computed based on 46 items from the Spanish test in 2020/2021 and 48 from the mathematics test in 2020/2021. In 2023, all 40 questions were included. There are 15 overlapping items in each subject across 2023 and 2020/2021.

The test questions included in the final scoring had acceptable and positive discrimination values, meaning that students of higher ability were more likely to answer correctly (Figures B.5-B.21). Further, the item characteristic curve fits were similar for students in both fifth and sixth grades and across assessment rounds (2020, 2021, and 2023), suggesting that comparing performance across grades and years reveals actual learning level differences.

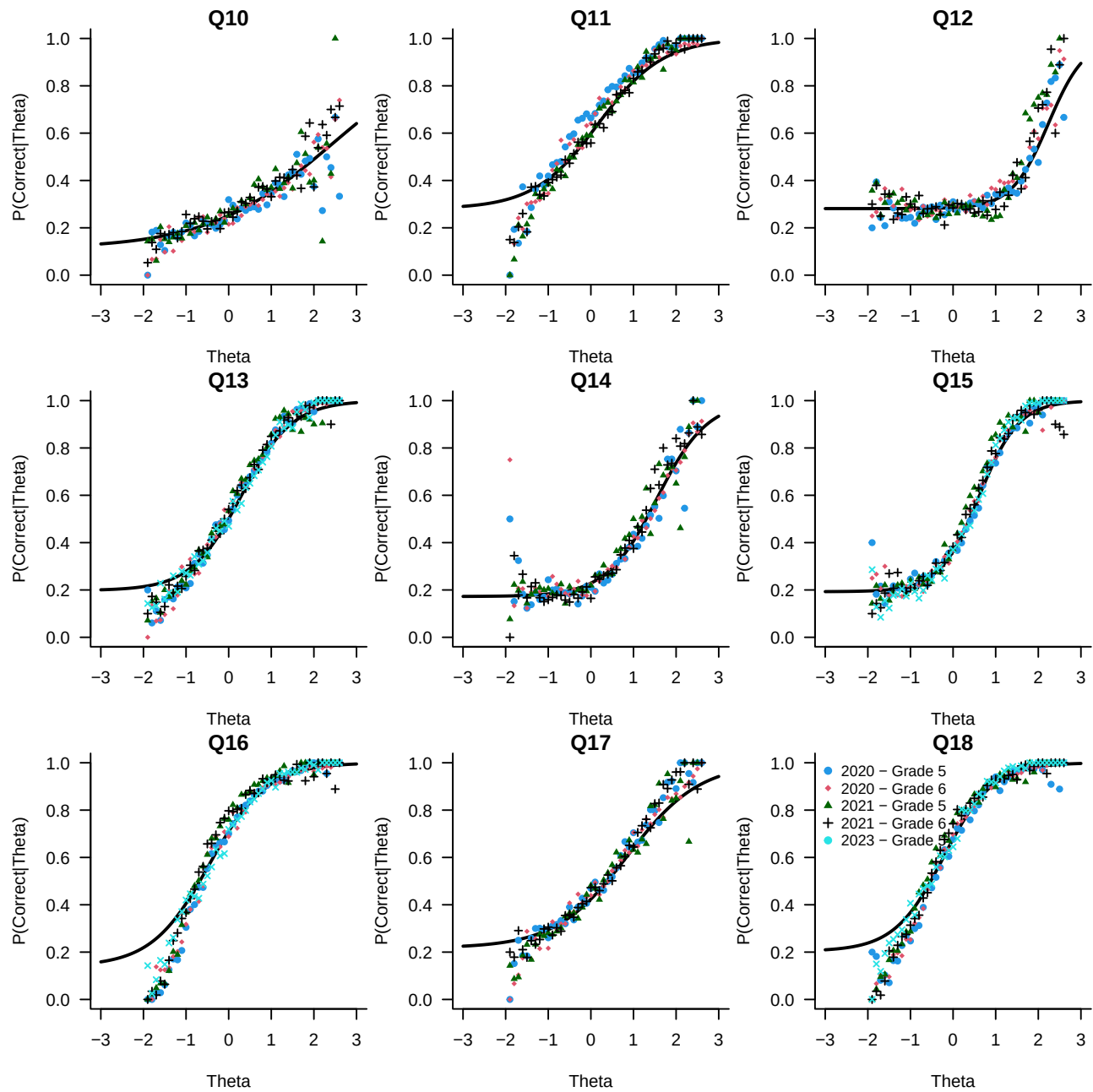
Finally, test reliability was estimated using (Cronbach, 1951)'s alpha, which measures how questions on the test are correlated. Test reliability for the 2020 and 2021 assessments was 0.84 in Spanish and 0.82 in mathematics. In 2023, reliability was 0.89 in Spanish and 0.87 in mathematics. Usually, reliability estimates of 0.7 or above are considered good in low-stake assessments (Nunnally, 1994; Bland & Altman, 1997; DeVellis, 2016; Ramírez & Valverde, 2021).

Figure B.5: Questions 1-9 - Spanish



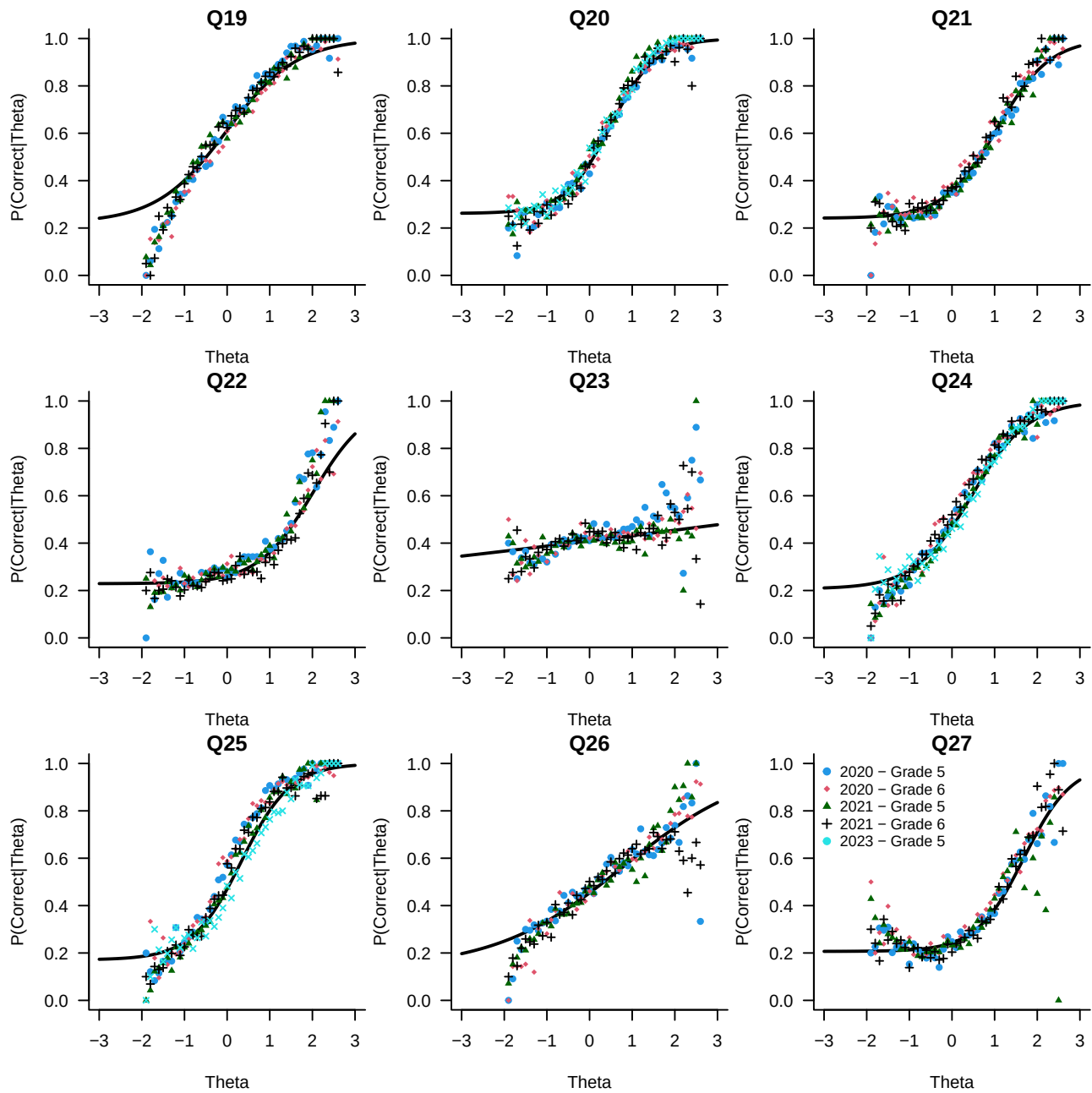
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.6: Questions 10-18 - Spanish



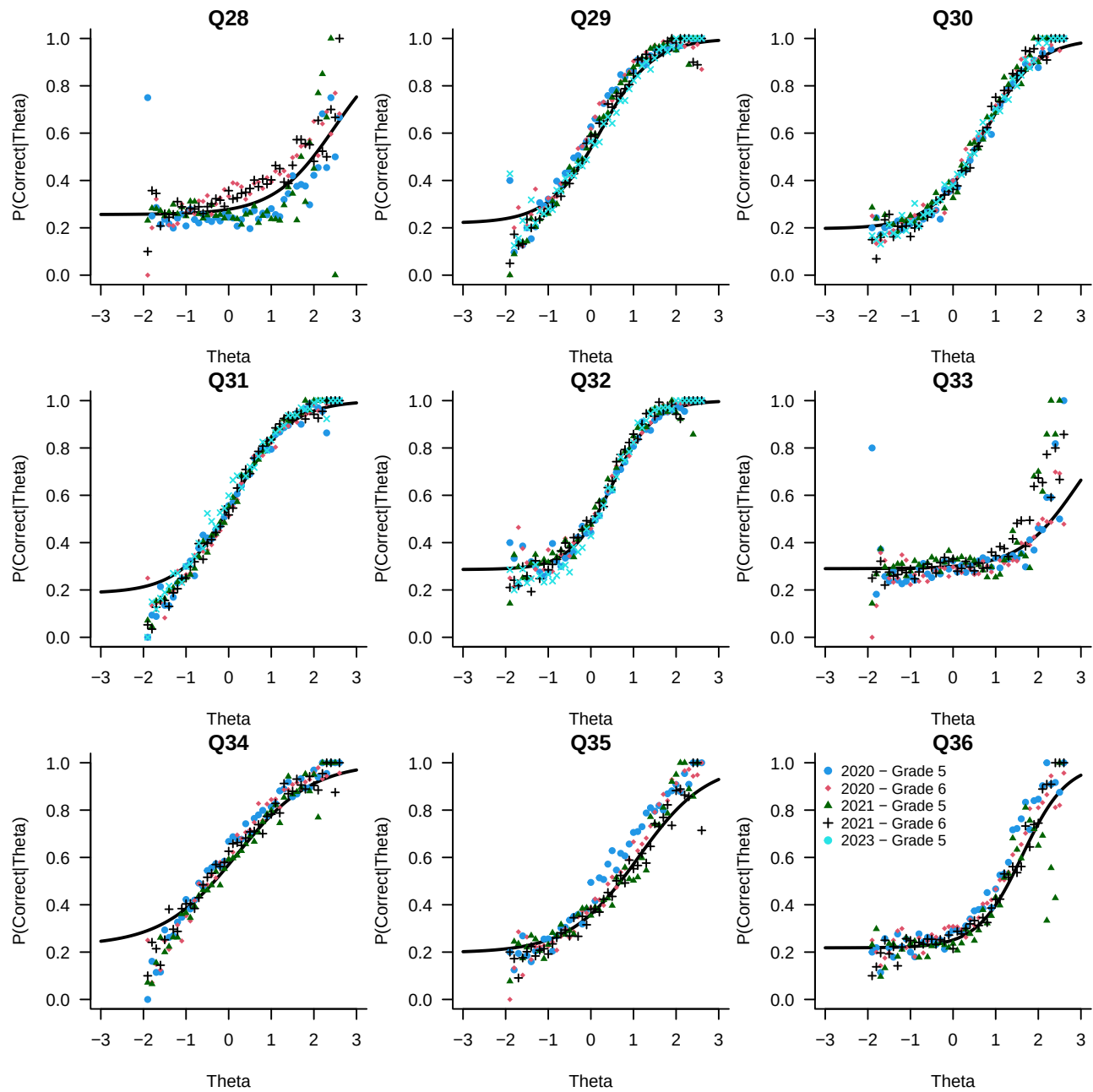
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.7: Questions 19-27 - Spanish



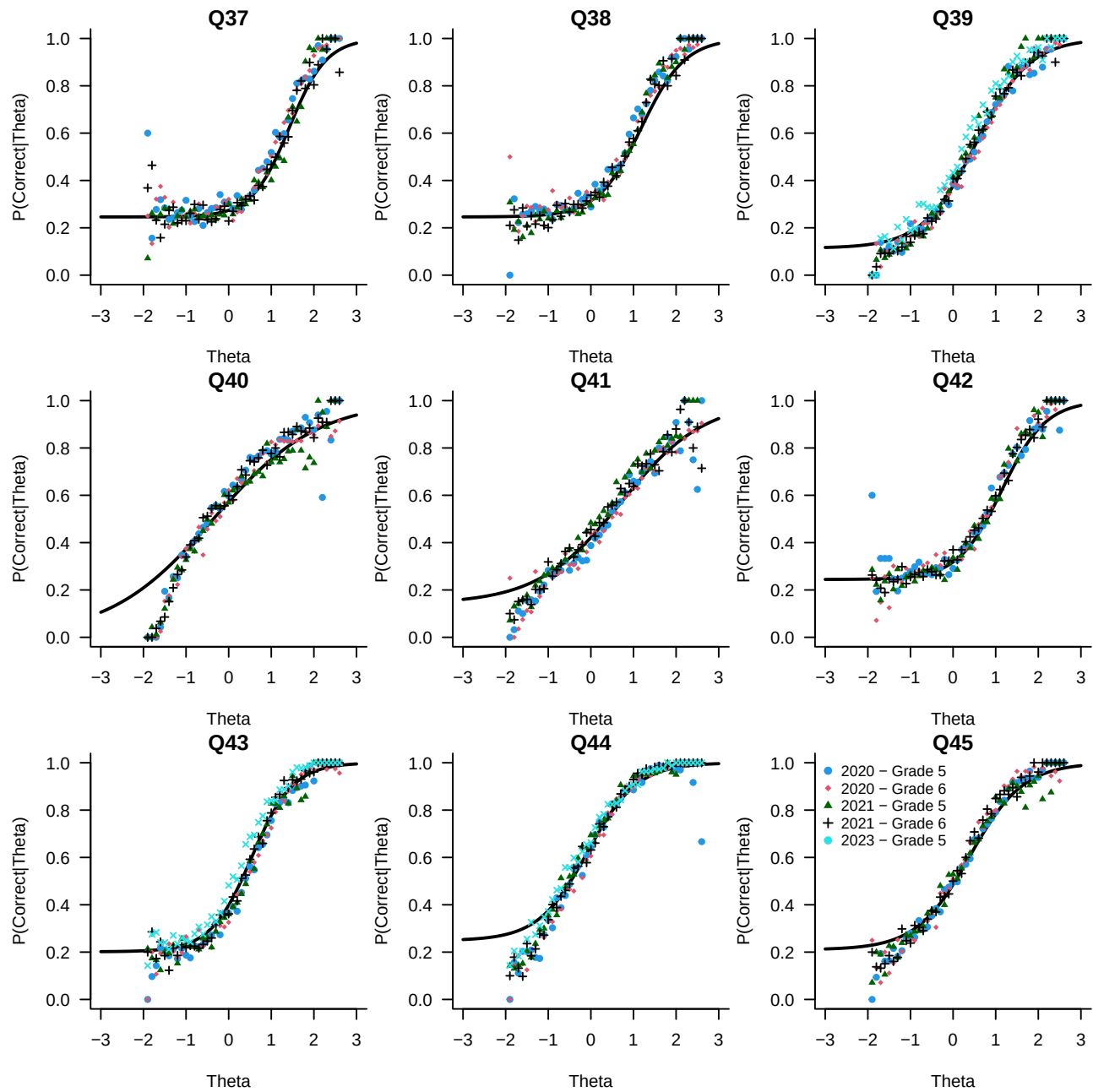
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.8: Questions 28-36 - Spanish



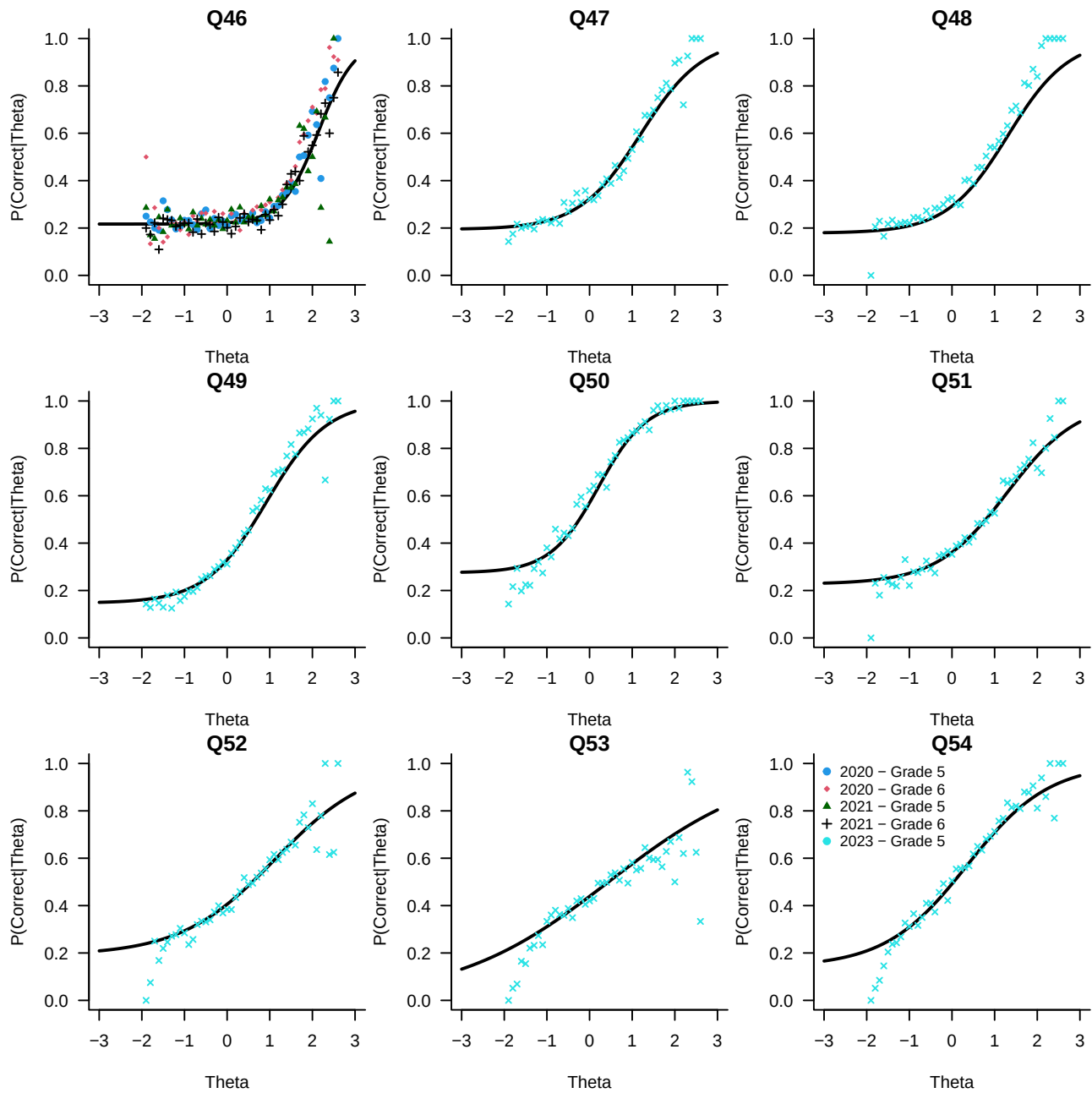
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.9: Questions 37-45 - Spanish



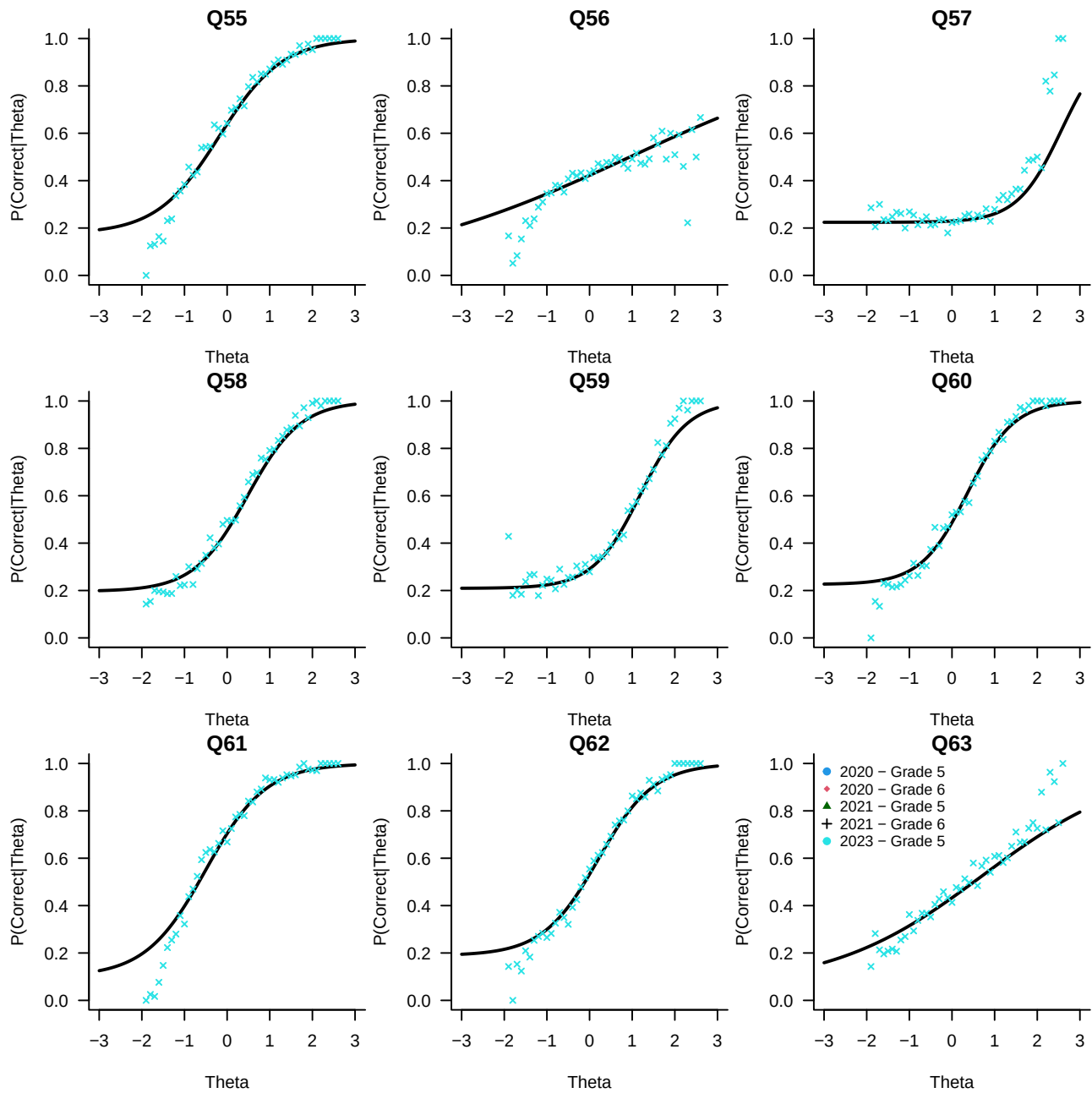
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.10: Question 46-54 - Spanish



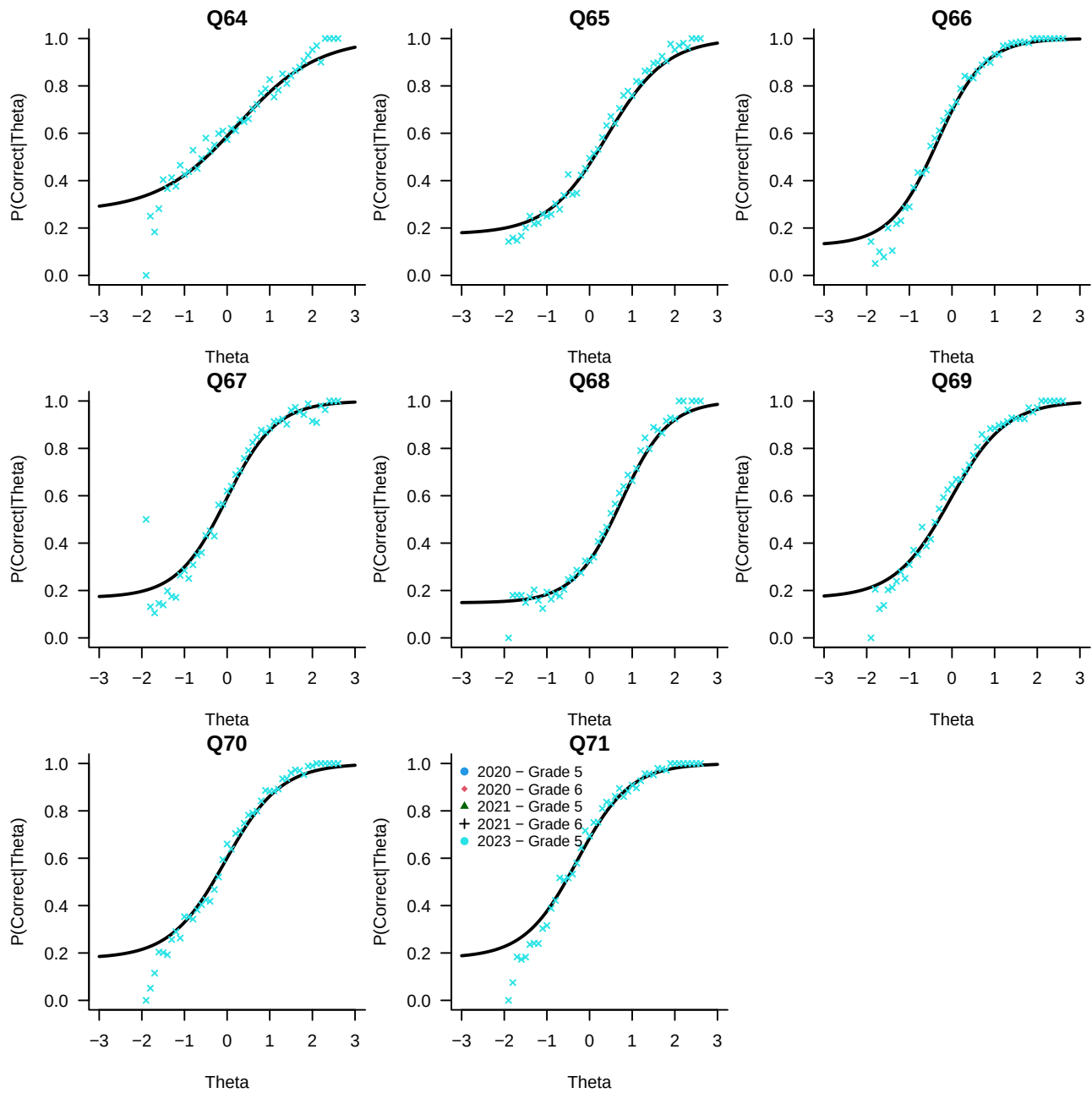
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.11: Question 55-63 - Spanish



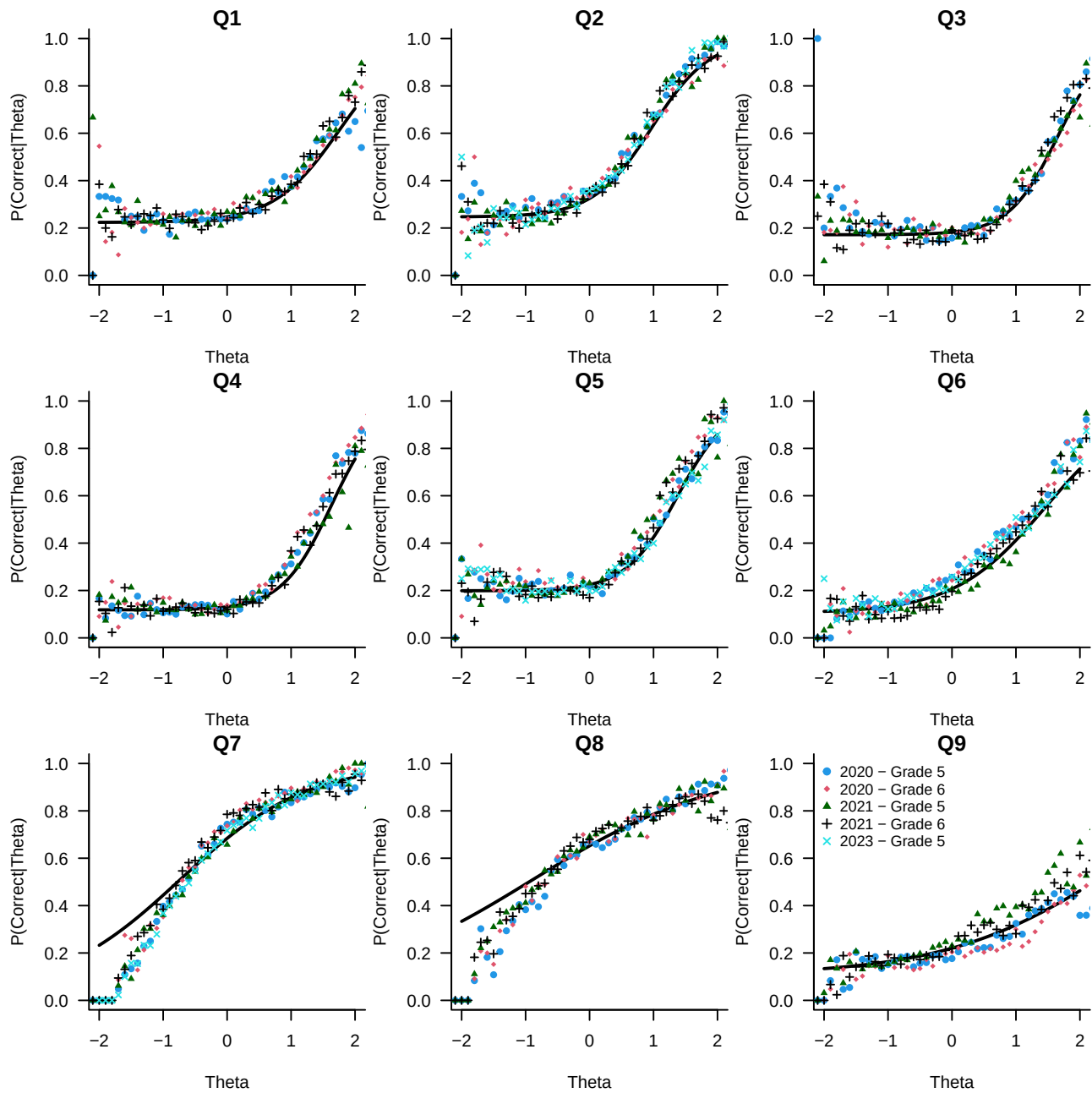
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.12: Question 64-71 - Spanish



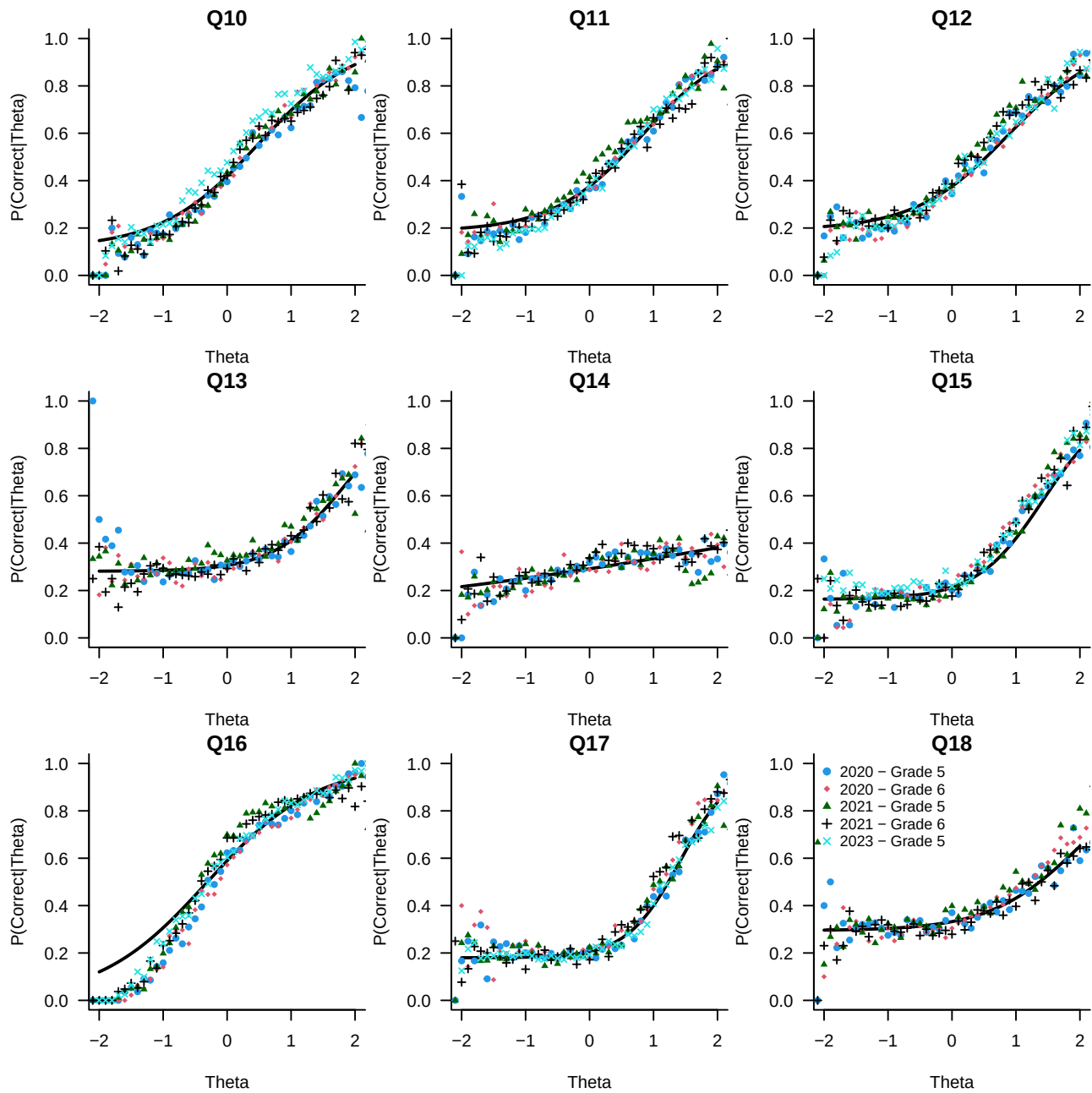
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.13: Questions 1-9 - Math



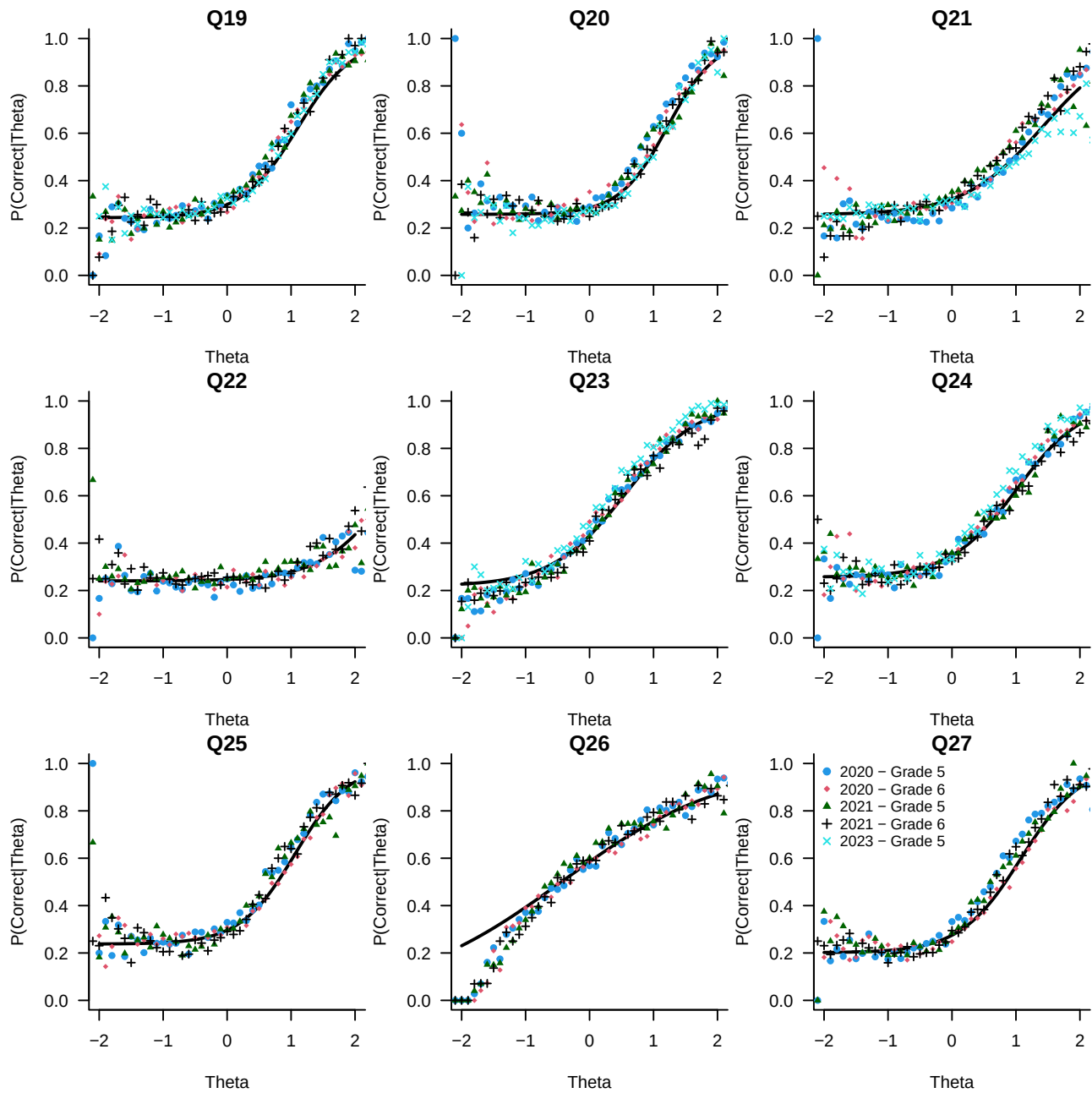
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.14: Questions 10-18 - Math



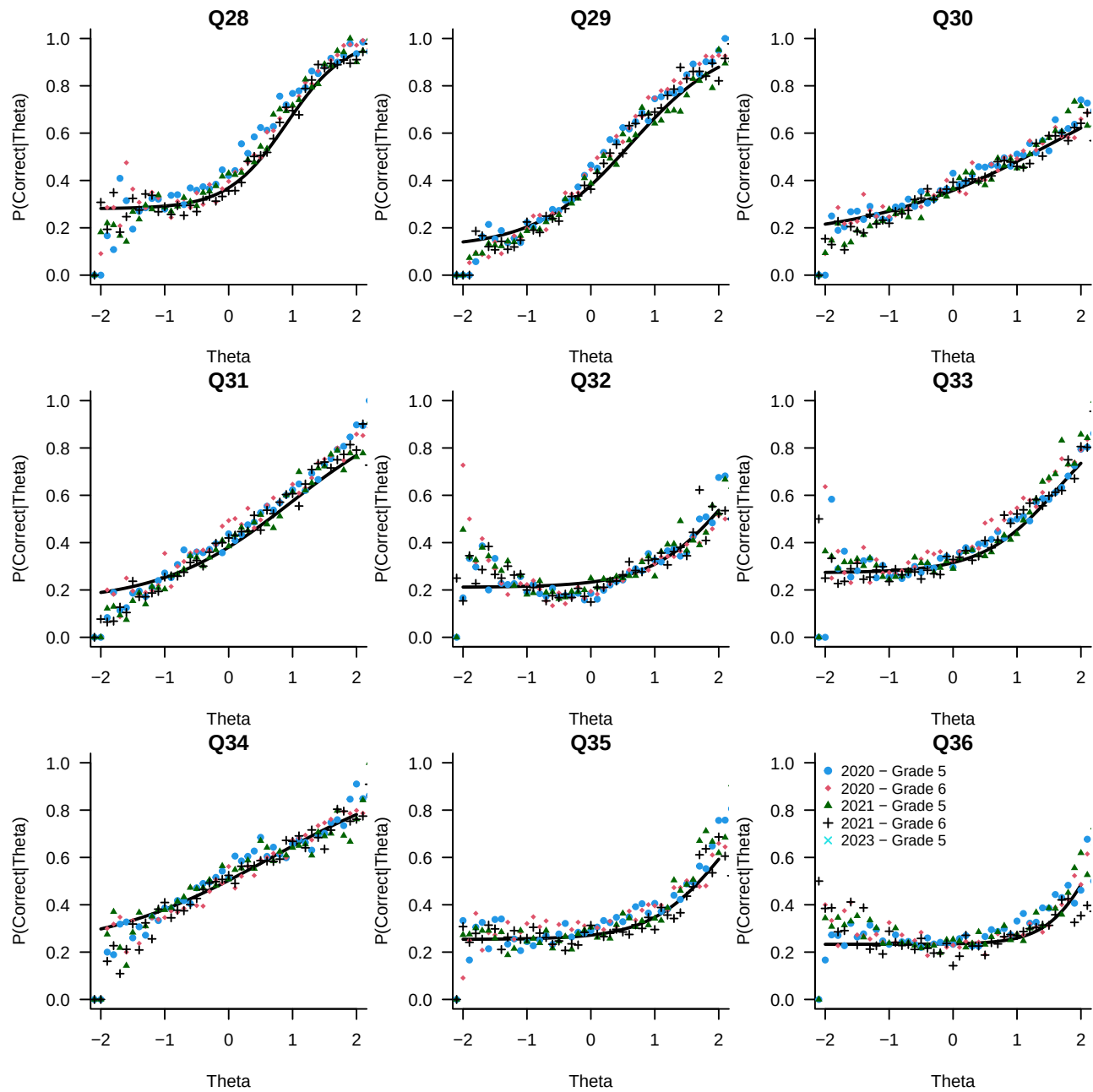
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.15: Questions 19-27 - Math



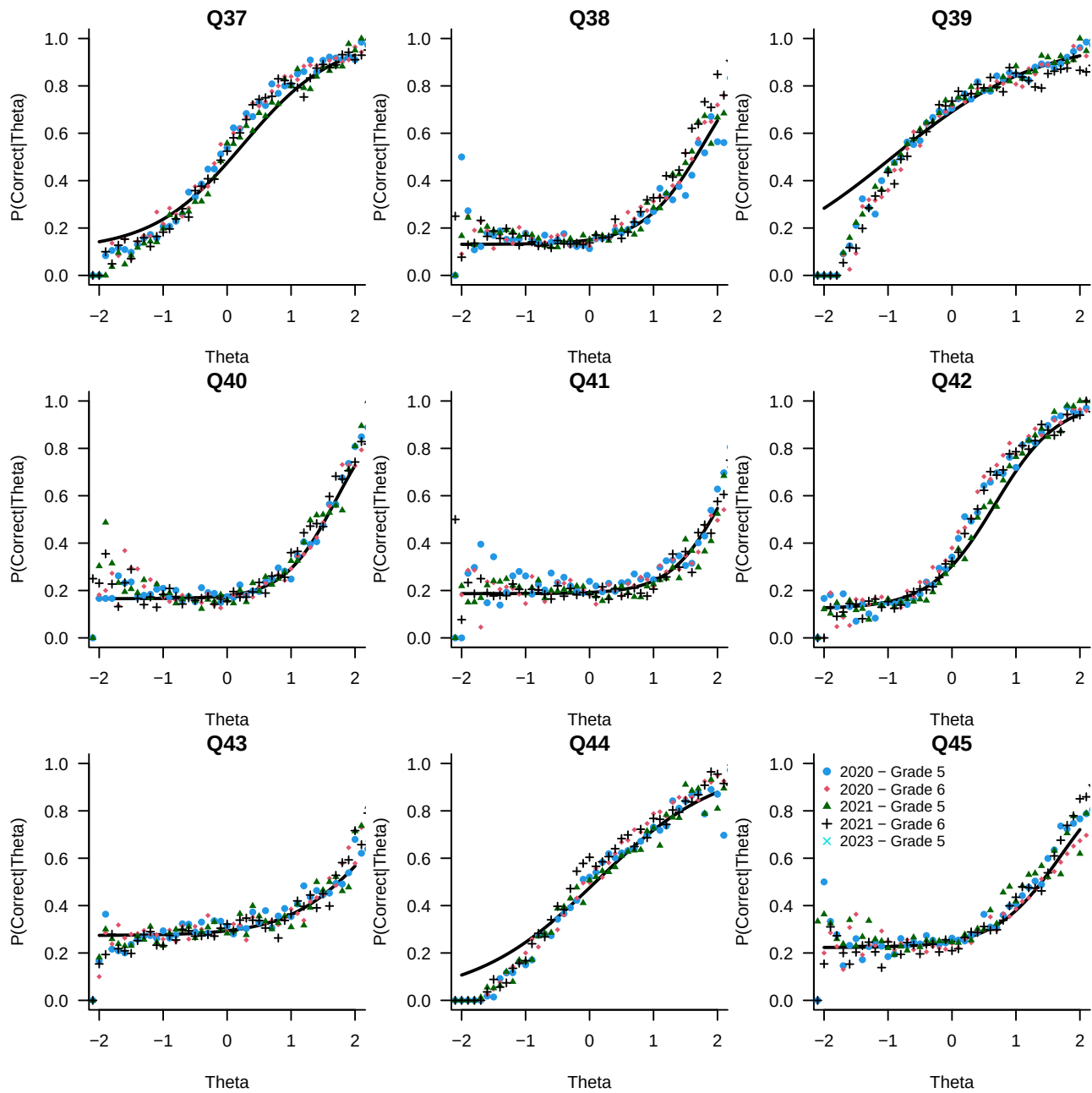
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.16: Questions 28-36 - Math



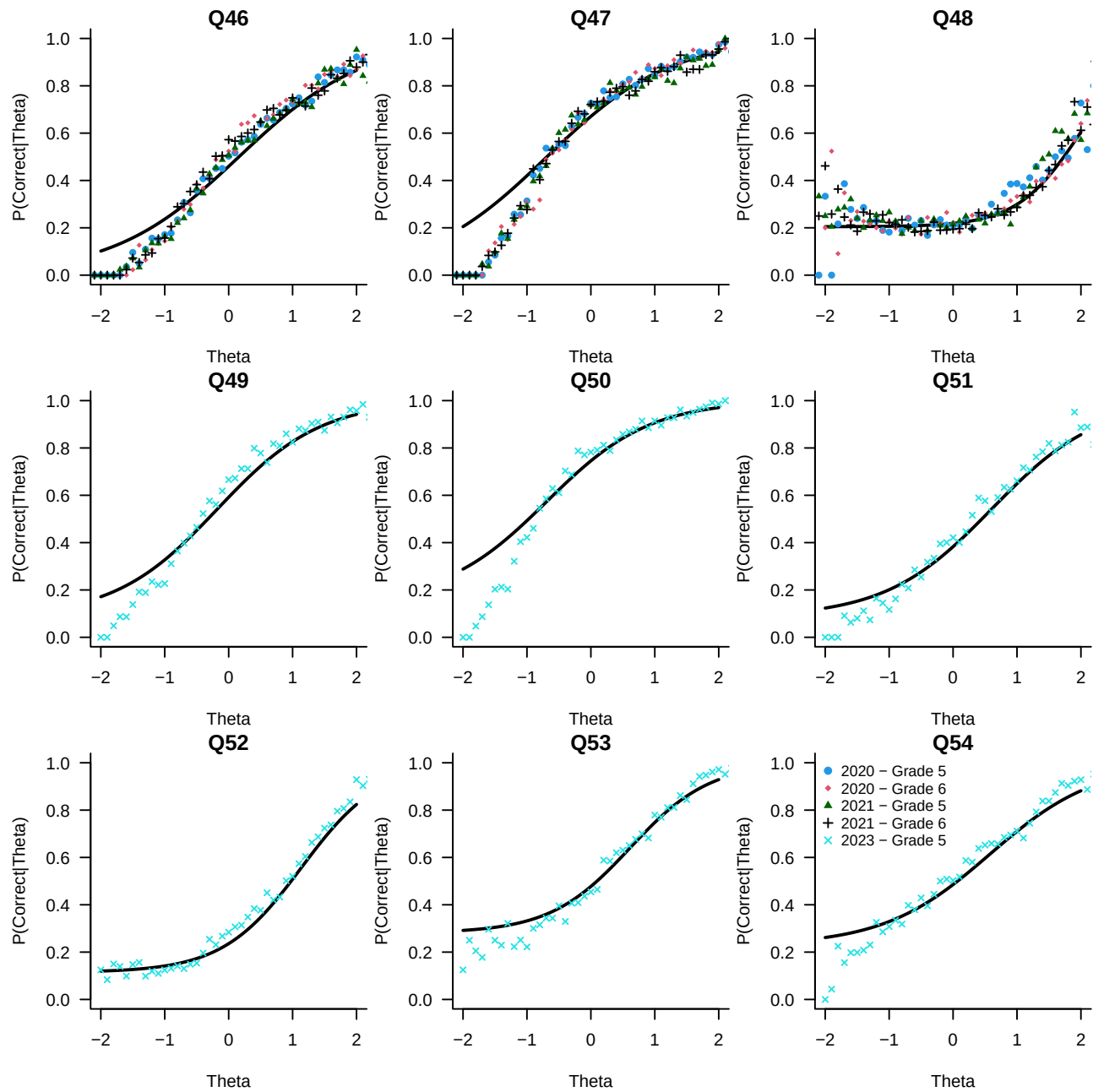
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.17: Questions 37-45 - Math



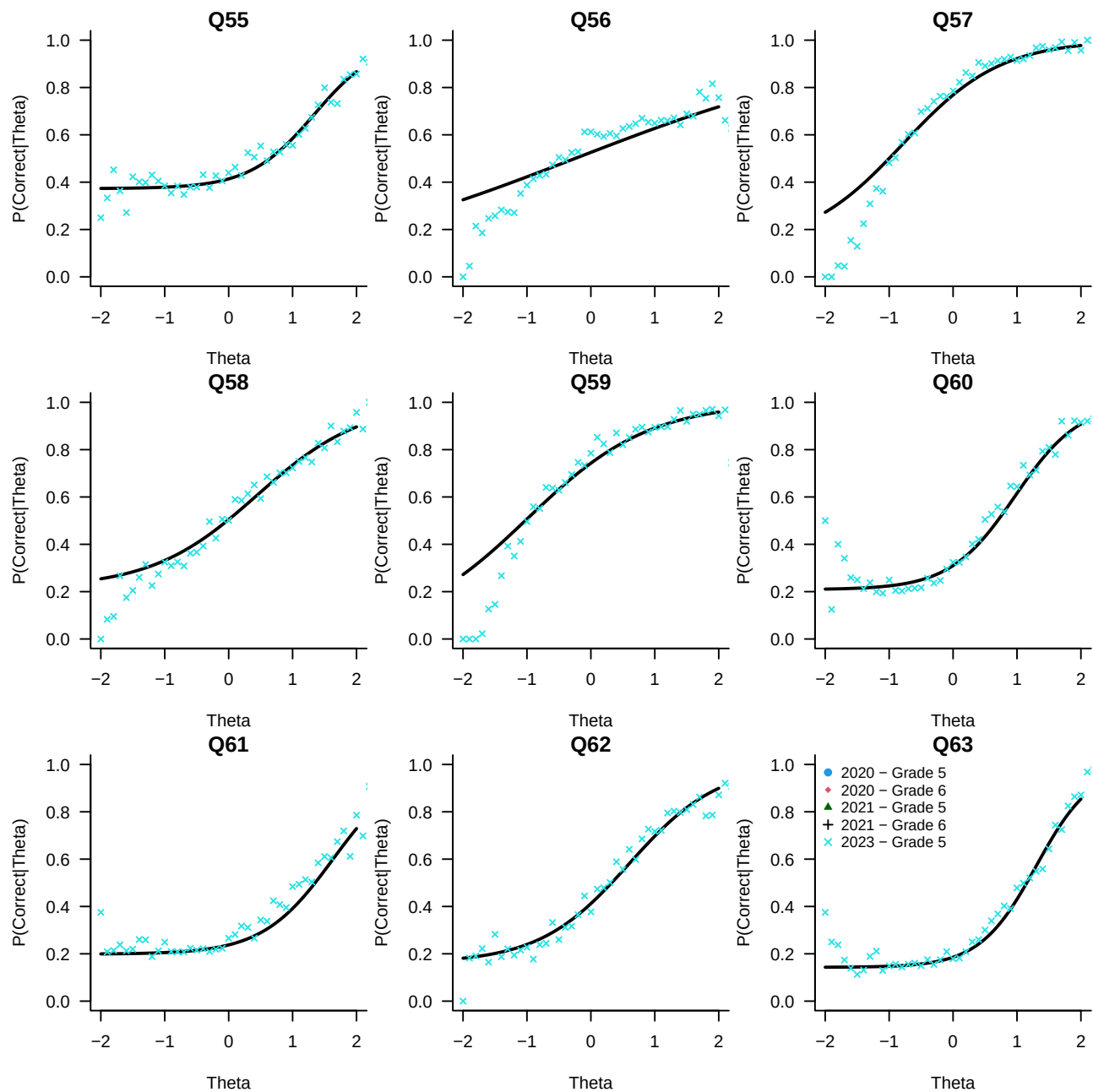
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.18: Questions 46-54 - Math



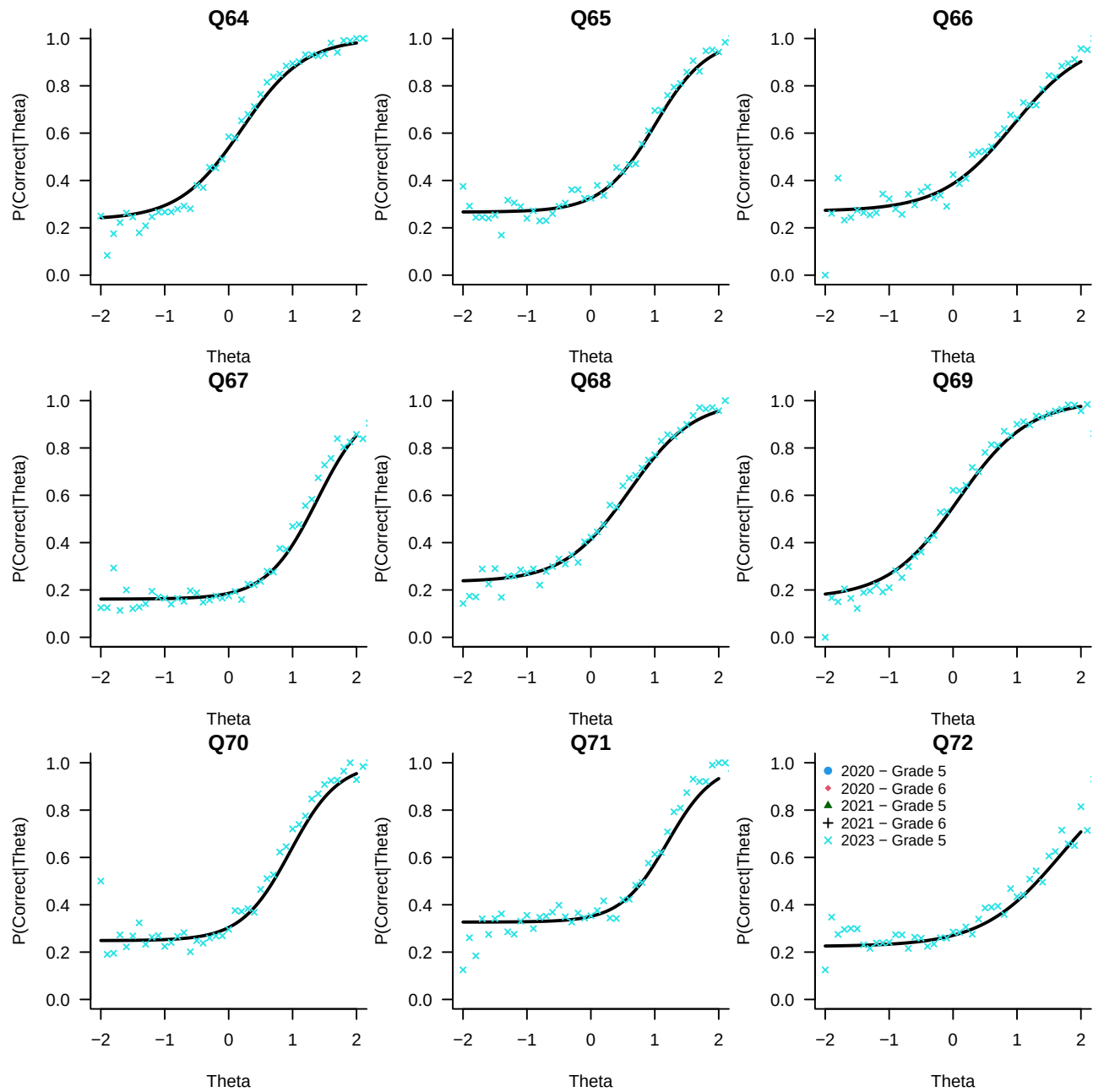
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.19: Questions 55-63 - Math



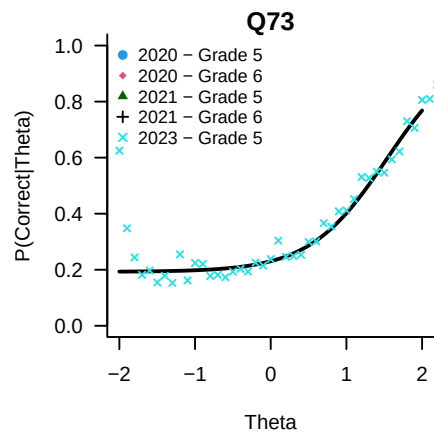
Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.20: Questions 64-72 - Math



Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.

Figure B.21: Questions 73 - Math



Note: This figure presents the likelihood students with different IRT scores answer different questions correctly, as well as the item characteristic curve for each question.